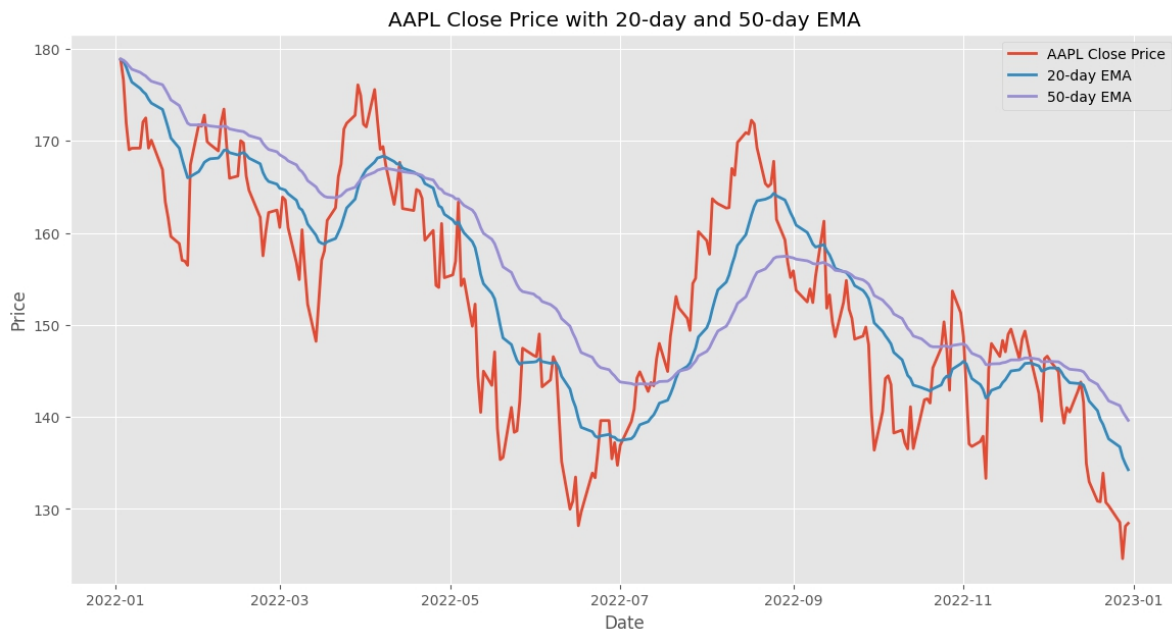




EMAs respond more quickly to price changes than SMAs, making them potentially more useful in fast-moving markets or for shorter-term trading strategies. However, this responsiveness also means they can generate more false signals during choppy market conditions.

Weighted Moving Average (WMA)

The Weighted Moving Average also assigns different weights to prices, typically in a linear fashion with the most recent price receiving the highest weight. Think of it as a compromise between the SMA (equal weights) and the EMA (exponentially declining weights).

```
1. def weighted_moving_average(data, window):
2.     weights = np.arange(1, window + 1)
3.     return data.rolling(window).apply(lambda x: np.sum(weights * x) / weights.sum())
4.
5. # Calculate 20-day Weighted Moving Average
6. apple['WMA20'] = weighted_moving_average(apple['Close'], 20)
7.
```